Smart Personal Finance Management System

A web application that helps users manage their personal finances by tracking income, expenses, budgets, savings goals, and recurring expenses. The system provides a secure platform where each user can view only their own financial information and understand their spending habits through reports and visual dashboards.

Problem Statement

People often struggle to monitor where their money is spent. They may record expenses manually or not track them at all, making budgeting and saving difficult. This project provides a centralized solution to organize financial data and generate meaningful insights.

Objectives

• Secure user registration and login.
• Record income and expenses.
• Categorize transactions.
• Set monthly budgets and savings goals.
• Display dashboards with charts and reports.
• Demonstrate core DBMS concepts using MongoDB.

Technology Stack

Frontend: React.js and Tailwind CSS. Backend: Node.js and Express.js. Database: MongoDB with Mongoose. Authentication: JWT and bcrypt.

Overall Workflow

Users register and log in to the application. After authentication, they can add income, record expenses, set budgets, and define savings goals. The backend validates requests, stores data in MongoDB, and performs calculations such as total expenses, remaining budget, and savings. The dashboard presents this information through charts, summaries, and reports.

Main Modules

1. Authentication – Register, login, profile.
2. Income & Expense Management – Add, edit, delete transactions.
3. Budget & Savings – Monthly budgets and savings goals.
4. Dashboard & Reports – Charts, analytics, transaction history, and financial summaries.

Dashboard

The dashboard displays total income, total expenses, remaining balance, savings, recent transactions, budget utilization, highest spending category, and interactive charts such as pie, bar, and line graphs.

DBMS Concepts

Separate collections for Users, Income, Expenses, Categories, Budgets, Goals, and Recurring Expenses. The project demonstrates CRUD operations, ObjectId relationships, indexing, aggregation pipelines, validation, and transactions.